

# **TIRESIAS: TImely REsistance Surmise In Additive manufacturing Samples**

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# 1 Step 1: Business Understanding

## 1.1 Business Goals

Additive Manufacturing (AM) refers to the process of fabricating objects through the sequential deposition of material layers using specialized machinery. This technology is widely adopted for rapid prototyping to evaluate the mechanical properties of components with complex geometries. The efficiency of prototype production facilitates extensive experimental campaigns involving mechanical stress tests, enabling the characterization of material properties through systematic data collection [1].

This study investigates the feasibility of reducing operational costs and accelerating experimental workflows by developing a virtual proxy of the physical test. This is particularly relevant in the production of high-precision complex components, where achieving defect-free specimens is challenging; ensuring a "fair" experimental environment is difficult, and preventing the destruction of such specimens could significantly reduce the cumulative waste generated during iterative manufacturing attempts. This framework aims to accurately predict material resistance within the shortest possible timeframe, potentially enabling the preservation and reuse of prototypes for subsequent testing.

Crucially, predictive accuracy must be balanced with timeliness. A model that yields results only upon the conclusion of a test offers limited utility, as the specimen would have already undergone irreversible deformation, rendering it unsuitable for reuse. Consequently, the primary objective of this research is to verify the feasibility of generating accurate resistance predictions with sufficient lead time.

## 1.2 Data Mining Goals

From a Data Mining perspective, this objective translates into achieving an average *earliness* of less than 50%. In this context, earliness refers to the model's capability to provide reliable outputs using only a partial sequence of the available experimental data. During the inference phase, each experiment is partitioned into *prefixes*—sub-sequences representing incremental steps of information. The model evaluates performance metrics across these prefixes, generating a curve that describes the evolution of accuracy as new data points are integrated. The *earliness* threshold is defined as the point on this curve beyond which the addition of further information does not yield a statistically significant improvement in predictive performance.

To achieve this, the proposed approach is structured into three primary phases:

1. Data preprocessing and transformation into a format suitable for temporal analysis.
2. Unsupervised learning to identify natural clusters and patterns within the experimental datasets, thereby isolating groups of experiments with distinct characteristics to develop specialized models for each category.
3. Supervised learning, specifically regression analysis, to predict the ultimate resistance of the analyzed specimens.

The project, complete with code, datasets, plots and results, can be accessed at the following GitHub repository: <https://github.com/AntonioCampanozzi/TIRESIAS>

### 1.3 Situation Assessment

A critical aspect of this study involves the evaluation of available computational and infrastructural resources. The ideal deployment scenario requires models with extremely low inference latency to avoid operational downtime while awaiting results. Furthermore, the underlying assumption is the presence of a high-speed, efficient data processing and transfer pipeline for streaming data, complemented by a dataset of sufficient scale to ensure robust model training.

## 2 Step 2: Data Understanding

### 2.1 Data Source and Dataset Composition

The dataset utilized in this research is publicly available through the Harvard Dataverse [2]. It comprises a total of 360 individual files, each representing a discrete experimental run. These experiments are categorized into two primary mechanical testing modalities: three-point bending and compression, with each category containing 180 files.

The experimental design is rigorously structured into 12 distinct groups per modality, with each group consisting of 15 experimental repetitions. This stratification accounts for four different thermoplastic materials—Polyamide (PA), Polycarbonate (PC), Polyethylene Terephthalate Glycol (PETG), and Polylactic Acid (PLA)—tested across three specific wall thicknesses (1 mm, 2 mm, and 3 mm).

Consequently, the dataset exhibits a highly balanced distribution, ensuring that each combination of material properties and geometric configurations is equitably represented. This inherent symmetry in the data distribution is critical for minimizing bias during the subsequent training and validation phases of the predictive models. The complete dataset is open-access and can be retrieved from the repository for replication and further analysis.

The datasets analyzed in this study are stored in **.TXT** files, structured as tabular sequences where data points are delimited by semicolons or tabs. Each file represents a discrete mechanical experiment—either a compression or a three-point bending test—performed on specimens fabricated via Additive Manufacturing.

In terms of **data volume**, each dataset is composed of a comprehensive header containing experimental metadata and material constants, followed by a high-resolution time-series of measurements. The number of attributes ranges from 7 to 12 depending on the test configuration, with thousands of records per file that capture the mechanical response from the initial pre-load to the point of structural failure.

The **feature identity** is categorized into categorical nominal identifiers, static discrete experimental parameters, and continuous quantitative variables:

- **Categorical Variables:** The primary identifier is the *Probenbezeichnung* (Specimen Designation), which encodes the material (e.g., Polyamide [PA] or Polylactic Acid [PLA]), the layer height (e.g., 0.1mm or 0.3mm), and the specific print orientation or batch.
- **Static Parameters and Targets (Header Data):** The header provides critical reference values that serve as the ground truth for regression. These include *Druckfestigkeit* or *Biegefestigkeit* (Compressive or Flexural Strength), *Druckmodul* or *Biegemodul* (Elastic Moduli), and specific limits such as the *0.2% Stauchgrenze*

(0.2% Offset Yield Strength). Additionally, operational constants like *Vorkraft* (Pre-load force), *Prüfgeschwindigkeit* (Test speed), and *Geschwindigkeit Biegemodul* (Modulus test speed) are recorded to ensure experimental replicability.

- **Quantitative Time series variables (Compression):** The sequential data for compression tests tracks *Prüfzeit* (Test Time) and *Traversenweg absolut* (Absolute Crosshead Travel). Crucially, it monitors *Standardweg* (Standard Displacement), *Stauchung* (Compression/Strain), and *Nominelle Stauchung* (Nominal Strain), providing a multi-dimensional view of how the material compacts under the applied *Standardkraft* (Standard Force).
- **Quantitative Time series variables (Three-Point Bending):** Bending experiments include unique metrics such as *Verformung* (Deflection/Deformation) and *Randfaserdehnung* (Edge Fiber Strain), which measures the elongation of the outermost specimen fibers. Other recorded features include *Standardweg* (Displacement), *Traversenweg* (Crosshead Travel), and *Biegespannung* (Flexural Stress), alongside the continuous recording of the *Standardkraft* (Applied Force).

## 2.2 Data Quality Analysis

To facilitate data quality assessment and model development, a custom parsing methodology was implemented to transform the raw **.TXT** files into a structured collection of Pandas DataFrames. During this process, feature names were standardized into English terminology to ensure technical clarity. Furthermore, the framework integrates experimental metadata as static, single-value variables and introduces new categorical features derived from the specimen designations, specifically identifying the material type and wall thickness for each experimental run.

Preliminary analyses were conducted to identify missing values ("nicht benakt") across both the input features and the potential regression targets. While no missing data points were recorded within the time-series measurements, metadata and target labels exhibited varying proportions of missingness, as summarized in Table 1.

| Feature Description                          | Count | Test type   |
|----------------------------------------------|-------|-------------|
| 0.2% Offset Yield Strength (Compressive)     | 180   | Compression |
| Total Strain at Maximum Compressive Strength | 180   | Compression |
| Compressive Stress at Maximum Strain         | 41    | Compression |
| Compressive Modulus                          | 29    | Compression |
| Compressive Modulus (via Hysteresis)         | 3     | Compression |
| Flexural Strength                            | 2     | Bending     |
| Deformation at Flexural Strength             | 2     | Bending     |

Table 1: Frequency of missing metadata across the experimental dataset.

## 2.3 Exploratory Data Analysis and Feature Correlation

Following the initial analysis, a more granular exploratory step was conducted to evaluate the correlation between input features. The primary objective was to determine the feasibility of horizontal dimensionality reduction by identifying and eliminating redundant features. Figures 1 and 2 present the correlation heatmaps for the bending and compression datasets, respectively, providing a visual justification for the feature selection process.

Simultaneously, force-deformation (stress-strain) curves were plotted to visualize the experimental trajectories. This step was fundamental in evaluating the natural dispersion or inherent grouping within the data.

To determine whether unsupervised learning algorithms were required or if the experiments could be effectively clustered using existing categorical attributes, the curves were analyzed by aggregating data based on material type, wall thickness, and a combination of both. By observing whether these groupings resulted in highly similar trajectories with low intra-class variance, I evaluated the potential for "domain-driven" clustering. For instance, if experiments partitioned by material exhibited distinct, non-overlapping clusters where individual curves followed nearly identical paths, the material type itself could serve as a primary clustering criterion. This visual and statistical validation ensures that the subsequent modeling phase accounts for the physical and geometric characteristics that most significantly influence the material's mechanical behavior, if they do.

Figures 3 and 4 illustrate the scenarios where the inter-curve distance is most pronounced, providing an insight into the worst-case variance encountered within each classification category.

## 3 Step 3: Data Preparation

### 3.1 Target Selection and Data Cleaning

As illustrated in Table 1, the compression datasets exhibit a significant lack of metadata, with *0.2% Offset Yield Strength* and *Total Strain at Maximum Compressive Strength* being unavailable across all experiments. Mapping these missing values was essential to determine the most viable target variable for regression. The selection of the target label was governed not only by availability but also by its relevance to the task and its potential relationship with the input features. It should be noted that direct correlation analysis between the static target and the raw dynamic time-series is mathematically inconsistent due to their differing dimensions.

Given the nature of the missing data, a deletion strategy was preferred over imputation. Since the target labels are missing entirely for certain subsets, imputing them from other experimental runs would introduce significant noise, as the subtle stochastic oscillations inherent in each mechanical test could lead to a biased representation of material failure. Consequently, *Compressive Stress at Maximum Strain* was selected as the target for the compression tests, even at the cost of dropping 41 datasets, while *Bending Strength* was chosen for the bending tests. Both metrics serve as robust proxies for the ultimate mechanical resistance of the specimen prior to structural failure.

The correlation analysis for the bending datasets, shown in Figure 1, reveals exceptionally high collinearity even in the worst-case scenario (Figure 1b). This justifies a drastic reduction of the feature space. I selected *Deformation (mm)* and *Force Applied*

( $N$ ) as the primary predictors; despite their strong relationship, they represent distinct physical domains (kinematics vs. kinetics) and exhibited a slightly lower correlation compared to other redundant metrics.

In contrast, the compression feature correlations (Figure 2) demonstrate a more diversified structure. Notably, *Deflection at Standard Load (%)* is significantly decorrelated from the remaining features. Metrics related to displacement, such as *Crosshead Displacement (mm)*, *Compression (%)*, and *Deformation (mm)*, show maximum collinearity even in the minimum case (Figure 2b). Since *Compression (%)* displayed a marginally lower correlation with the applied force than other displacement metrics, it was retained as the representative kinematic feature. Therefore, the final feature set for the compression models consists of *Compression (%)*, *Deflection at Standard Load (%)*, and *Force Applied (N)*.

### 3.2 Data Construction

In this study, the introduction of composite features was intentionally avoided. Given the high granularity and volume of the time-series data, maintaining a lower dimensionality and limited multivariety was prioritized to prevent computational bottlenecks and overfitting. Instead, two primary data transformations were implemented to prepare the dataset for unsupervised and supervised tasks. For the clustering phase, the DataFrames were transformed into a format compatible with the `tslearn` library. Specifically, the data was reshaped into a three-dimensional tensor with the structure  $(n_{samples}, n_{timestamps}, n_{features})$ . To ensure that the clustering algorithms evaluate the characteristic shape of the mechanical trajectories rather than their absolute magnitudes, I applied Z-score normalization (Mean-Variance scaling). This standardization process centers each feature’s distribution around a mean of zero with unit variance, ensuring that kinematic and kinetic variables contribute equitably to the distance metrics.

To support the early prediction objective, a second transformation was applied to create a "prefix dataset." The original time-series were decomposed into batches of partial sequences (prefixes) of random length for each sequence of the batch for training set, to add variety during the learning phase, and equal lengths for test set, to make possible the computation of the earliness. Since individual experiments often vary in duration, shorter sequences were extended using *zero-padding* to reach a uniform length, as illustrated in Figure 5.

During the training of the supervised models, a masking mechanism was implemented to ensure that the padded values are ignored by the loss function and the gradient updates. This approach allows the model to process fixed-length tensors within each batch while preserving the temporal integrity of the underlying experimental data.

### 3.3 Data Format

Due to the substantial memory footprint of the high quantity of temporal sequences, the finalized datasets are not persisted as physical files on disk. Instead, they are maintained as in-memory objects within the execution environment. This architectural choice was made to optimize resource utilization, as storing such a vast number of multi-dimensional arrays would result in a large number of files, difficult to browse.

For transparency and structural audit purposes, only two representative **.CSV** files have been exported. These files provide a tangible reference for the post-parsing data architecture, allowing for a manual inspection without the need for full-scale data persistence.

## 4 Step 4: Modeling

### 4.1 Model Techniques

As previously outlined, the proposed methodology integrates an unsupervised learning phase prior to supervised modeling. Preliminary visual inspections (Figures 3 and 4) indicate that partitioning data based solely on material type or wall thickness results in high intra-group variance. While combining these two attributes could theoretically isolate more cohesive groups, such a strategy would yield 12 distinct categories. This granularity poses a significant risk of over-specialization, leading to models with poor generalization capabilities due to the limited number of training samples available per cluster.

To address this, I employed a **K-Medoids** clustering algorithm. Unlike K-Means, K-Medoids selects actual experimental sequences as centroids, ensuring that the representative trends are physically grounded.

Inspired by the *CARROT* [3] framework, **Dynamic Time Warping (DTW)** was utilized as the distance metric to compute the dissimilarity matrix. DTW is particularly effective for mechanical stress-strain data, as it accounts for temporal shifts and varying speeds in the material’s response.

For the supervised regression phase, a **Long Short-Term Memory (LSTM)** network was selected as the core architecture. LSTMs remain a robust and reliable standard for processing time-series data, particularly in scenarios where capturing long-term temporal dependencies is critical for accurate early prediction.

The data preparation phase for the LSTM involved a feature scaling process applied directly to the batches, as the input variables exhibited significantly different magnitudes. To ensure robust evaluation, each cluster-specific sub-dataset was partitioned into training, validation, and test sets. The validation set was specifically utilized to implement an *early stopping* criterion, preventing the training process from continuing beyond the point of convergence and thereby mitigating the risk of overfitting.

The models were configured with a high number of epochs to accommodate potentially slow convergence patterns. A moderate learning rate was selected to maintain gradient stability; a rate that is too low could exacerbate signal attenuation in batches containing longer prefixes, ultimately compromising the network’s predictive capacity. A *patience* value of 15 epochs was established for the early stopping callback. This parameter provided a consistent tolerance threshold, allowing the model sufficient opportunity to escape local minima or plateau phases before terminating the training process prematurely.

This balance ensured that the optimization was extensive enough to capture complex temporal dependencies while still preventing unnecessary computational expenditure once the validation loss ceased to improve.

The LSTM network was configured with an *hidden size* of 128 units, which proved to be an effective trade-off between model capacity and the risk of overfitting, given the complexity of the experimental data. To manage the computational overhead and facilitate the learning of long-term dependencies, a temporal downsampling (decimation) strategy was implemented, retaining one data point for every ten original observations. Without this reduction, the raw sequences exhibited excessive length, leading to vanishing gradient issues and prohibitive training times that hindered the model’s ability to capture the overall mechanical trend.

For the optimization objective, the **Huber Loss** was selected as the loss function. This choice is motivated by its superior robustness compared to Mean Squared Error (MSE) and Mean Absolute Error (MAE). The Huber Loss effectively combines the advantages of both: it acts quadratically for small errors (like MSE), ensuring smoothness near the minimum, and linearly for large errors (like MAE), which reduces the influence of outliers. In the context of mechanical testing, where sensor noise or unexpected material failure can produce anomalous data points, the Huber Loss ensures more stable convergence by preventing large residuals from dominating the gradient updates.

## 4.2 Model Assessment

The number of clusters generated by the K-Medoids algorithm—four for bending and three for compression—is significantly lower than the twelve categories that would result from a combined material and thickness partitioning. However, the sample distribution across these clusters is highly non-uniform. As illustrated in Figure 6, both experimental modalities contain a "minority" cluster with substantially fewer observations. This imbalance suggests the presence of an anomalous subset of experiments exhibiting non-standard mechanical behaviors that differ from the general population trends.

To determine the resulting average earliness, a sequence was constructed where each element represents the error metric of a specific batch. This sequence was processed using a *moving average* to smooth the signal, thereby mitigating the impact of noisy fluctuations and anomalous peaks while preserving the sample size. The specific point of **Early Prediction** was then identified by evaluating the relative "Gain" between consecutive batches.

The Gain is defined as the reduction in error compared to the previous batch, normalized against the total potential gain. The early prediction point is triggered when this normalized gain reaches a threshold of 0.9. This indicates that the batch has captured 90% of the maximum possible predictive improvement, representing the optimal moment to terminate the experiment with high confidence in the result. The performance metrics for the LSTM models across these clusters are summarized in Table 2.

| Modality           | Cluster    | Test Loss (Huber) | Mean Earliness | Median Earliness |
|--------------------|------------|-------------------|----------------|------------------|
| <b>Bending</b>     | Cluster 0  | 0.0938            | 70.59%         | 82.35%           |
|                    | Cluster 1  | 0.7978            | 55.56%         | 51.85%           |
|                    | Cluster 2  | 0.0490            | 78.69%         | 81.97%           |
|                    | Cluster 3  | 0.1691            | <b>4.35%</b>   | <b>4.35%</b>     |
|                    | No Cluster | 0.1512            | 83.61%         | 86.89%           |
| <b>Compression</b> | Cluster 0  | 0.1451            | 92.22%         | 93.00%           |
|                    | Cluster 1  | 0.1383            | 82.40%         | 82.40%           |
|                    | Cluster 2  | 1.7349            | <b>42.41%</b>  | <b>42.41%</b>    |
|                    | No Cluster | 0.0726            | <b>47.40%</b>  | <b>47.40%</b>    |

Table 2: LSTM Model Performance: Acceptable earliness result are highlited in bold

While the LSTM architecture demonstrated a nominal ability to internalize the data patterns—as evidenced by the convergence of the Huber Loss on the test set—it failed to achieve a satisfactory *earliness* threshold. Specifically, the predictive accuracy only reached acceptable levels toward the conclusion of the experiments, indicating that the model was unable to generate reliable forecasts based on the initial temporal prefixes. The only notable exception is in the smallest clusters of both groups. However, an earliness so low in the Bending can be surely considered as noise, symptom of n irregular structure of test. Consequently, the LSTM has performed accettably only on 2 out of 9 groups,

## 5 Step 5: Evaluation

The technical results obtained through the LSTM approach do not meet the primary business objective, which required accurate material resistance predictions prior to the midpoint of the experimental run.

### 5.1 Evaluate Results

The evaluation confirms that while the LSTM model is effective for full-sequence regression, its utility for early-warning detection is limited in this specific experimental setup. The average earliness consistently exceeded the 50% threshold, failing to provide sufficient lead time for specimen preservation.

### 5.2 Review Process

In accordance with the iterative nature of the CRISP-DM methodology, the failure to meet the business goals necessitated a revision of the modeling strategy. Consequently, a second supervised learning approach was explored, shifting toward classical machine learning architectures. This decision was driven by the rationale that further increasing model complexity would likely exacerbate overfitting, especially in data-constrained scenarios. Furthermore, the elevated computational and maintenance overhead associated with more complex deep learning structures could no longer be justified by the primary objective of reducing specimen waste. Adopting a more efficient, tabular-based model ensures a better balance between predictive performance and the practical sustainability

of the workflow. The following sections detail the architectural modifications and the subsequent improvements in predictive performance.

## 6 Second Iteration

### 6.1 Data Construction and Feature Engineering

To align with the requirements of a tabular machine learning approach, a new transformation was applied to the parsed data. In this iteration, each temporal prefix is summarized into a single observation vector. Instead of utilizing raw sequential values, each prefix is represented by a set of statistical descriptors for each input feature that capture its fundamental dynamics:

- Mean and Standard Deviation (to capture central tendency and dispersion).
- Skewness and Kurtosis (to describe the shape and tail-heaviness of the stress distribution).
- Last observed value (to provide the most recent state of the experiment).
- Delta (the difference between the first and last elements, representing the overall trend).

Consequently, a single time-series is represented in the dataset by a number of rows equal to the number of extractable prefixes. Unlike the previous deep learning approach—where prefixes were generated on-the-fly via `__getitem__` to save memory—the summarized dataframes were persisted as **.CSV** files. This strategy is computationally more efficient, as it avoids the repetitive and high-RAM cost of statistical extraction during each execution.

### 6.2 Modeling Technique

The **XGBoost** (Extreme Gradient Boosting) algorithm was selected for this iteration due to its high efficiency and its proven performance on tabular benchmarks, often rivaling deep learning architectures. The model was configured with 1,000 estimators to ensure sufficient capacity to capture the diverse mechanical behaviors across the dataset without underfitting. A learning rate of 0.05 was chosen to facilitate a stable and gradual optimization process, reducing the risk of overshooting the global minimum. Additionally, a *max\_depth* of 5 was implemented to control model complexity; this depth allows the algorithm to learn intricate non-linear interactions between features while maintaining strong generalization capabilities and preventing overfitting.

A *hold-out* split was utilized for validation. For performance evaluation, Mean Squared Error (MSE) and R-squared ( $R^2$ ) were selected: MAE provides a direct measure of prediction error magnitude, while  $R^2$  indicates the proportion of variance captured by the model, offering a normalized metric of its explanatory power.

### 6.3 Model Assessment

The strategy for evaluating *earliness* remains consistent with the previous phase, though the tabular nature of XGBoost allows for a more granular assessment. Unlike the LSTM, where earliness was constrained by batch-level evaluation, this architecture permits the calculation of earliness for each individual sequence, which can then be aggregated. This

provides a more precise representation of the model’s predictive timing across different experimental conditions.

The results, as summarized in Table 3, demonstrate a significant improvement in predictive performance compared to the first iteration.

| Modality           | Cluster    | MAE    | $R^2$   | Mean Earl.    | Med. Earl.    | Std. Earl. |
|--------------------|------------|--------|---------|---------------|---------------|------------|
| <b>Bending</b>     | Cluster 0  | 2.8232 | 0.7726  | 48.24%        | 52.94%        | 32.50%     |
|                    | Cluster 1  | 3.5062 | 0.0888  | 72.10%        | 77.61%        | 28.33%     |
|                    | Cluster 2  | 3.7620 | 0.8517  | <b>46.17%</b> | <b>39.27%</b> | 31.44%     |
|                    | Cluster 3  | 3.1762 | -10.236 | 68.13%        | 53.07%        | 22.23%     |
|                    | No Cluster | 2.9949 | 0.9379  | <b>44.42%</b> | 50.83%        | 23.14%     |
| <b>Compression</b> | Cluster 0  | 3.3904 | 0.2983  | 45.83%        | 37.23%        | 25.20%     |
|                    | Cluster 1  | 2.2167 | 0.3891  | <b>25.27%</b> | <b>17.23%</b> | 22.73%     |
|                    | Cluster 2  | 7.9891 | -0.2079 | <b>33.16%</b> | <b>25.00%</b> | 33.01%     |
|                    | No Cluster | 2.5813 | 0.8758  | <b>20.30%</b> | <b>15.24%</b> | 14.38%     |

Table 3: XGBoost Model Performance: Acceptable earliness result are highlighted in bold

The experimental results demonstrate that the XGBoost-based approach significantly outperforms the LSTM architecture, with 5 out of 9 sub-groups successfully meeting the predefined earliness threshold. However, the findings also highlight critical limitations of the clustering strategy within this framework. As observed in the performance of the unified dataset, training a single, consolidated model leads to a substantial improvement in  $R^2$  values and a marked reduction in the standard deviation of the earliness. This indicates that a global model is more robust and consistent than the specialized sub-models.

This outcome is particularly significant from an engineering perspective: it proves that a streamlined methodology based on statistical feature engineering can surpass a complex deep learning architecture in both predictive accuracy and operational manageability.

## 6.4 Evaluation of Results

The second iteration successfully achieved the primary business objectives. When evaluated through the clustering lens, the majority of the sub-groups reached the targets; when considering the non-clustered variant, the objectives were fully met. These results confirm that it is possible to obtain highly accurate material resistance predictions with sufficient lead time to preserve specimens, validating the feasibility of the early prediction system.

## 6.5 Review and Future Work

In accordance with the iterative nature of the CRISP-DM framework, several avenues for future development have been identified. Future research should explore the application of **Feed-Forward Neural Networks (FFNN)** on the summarized tabular data to determine if a non-linear global model can further enhance performance. Additionally, a systematic hyperparameter fine-tuning of the XGBoost regressor could be performed

to establish whether the current predictive threshold represents a mathematical bottleneck inherent in the data or if further improvements in earliness are achievable through optimization.

## 7 Third Iteration

### 7.1 Business & Data Mining Goals

The imposition of a static early prediction threshold (50% of the total sequence) fails to account for the diverse nature of the tests, as it assumes a universal "point of no return" across all experiments. Consequently, the refined business goal is to predict the maximum stress supported by a specimen with sufficient lead time relative to the specific failure threshold characteristic of its respective cluster.

In terms of **data mining objectives**, the task involves evaluating the model's performance both at the cluster level and across the entire dataset. The results are considered satisfactory if the length of the prefixes—where the regression error remains sufficiently low compared to the ground truth—is significantly shorter than the reference thresholds. Specifically, for Cluster-level evaluation, the early prediction point, expressed as a percentage of the series must show a significant negative deviation from the specific cluster threshold, also expressed as percentage for scale unification. On the complete dataset the early prediction percentage must be compared against the mean of the thresholds (if the distribution is centered) or the median threshold across all clusters.

The data mining goal is achieved if this temporal advantage is consistently observed across the majority of clusters and maintained in the comprehensive testing phase, thereby validating the model's predictive efficiency.

### 7.2 Exploratory Data Analysis

The initial phase of the analysis focused on examining the force curves within each cluster to validate the use of medoids as representative prototypes. This approach was evaluated as a potentially robust simplification compared to the computationally expensive task of identifying individual thresholds for every curve.

The analysis aimed to verify the intra-cluster morphological similarity of the force profiles. Figures 8 and 7 demonstrate highly consistent trends within each cluster, notwithstanding variations in magnitude and time-series translation. These discrepancies were effectively addressed during the clustering phase via Dynamic Time Warping (DTW); furthermore, the application of feature scaling has mitigated the impact of varying force magnitudes across different series.

Consequently, the results support the decision to utilize cluster medoids as reference points. This is particularly effective when the threshold is defined as a percentage of the total sequence length rather than an absolute temporal index, thereby normalizing the variations in sequence duration across the dataset.

### 7.3 Threshold Definition

The visual analysis of force trends across all medoids presented in images indicates that distinct criteria must be established for Bending and Compression tests.

In the case of **Bending**, identifying the failure point is straightforward: it corresponds to the peak of the curve. Following this maximum value, a decrease in force—whether abrupt or gradual—signals that the sample’s structural integrity has been compromised and it can no longer sustain additional load.

Regarding **Compression**, the profiles typically exhibit two distinct peaks. For the purposes of this study, the *first peak* has been identified as the primary failure threshold. It is hypothesized that subsequent increases in force may result from the progressive densification or secondary structural resistance of the internal honeycomb/cellular lattice after the initial failure. However, a definitive characterization of this phenomenon would require further validation from domain experts.

Table 4 summarizes the failure thresholds across the entire dataset, reporting both standard deviation and median values. The remarkably low standard deviation observed justifies the use of the mean threshold as a reliable and stable representative metric for the global dataset in the subsequent analytical steps.

| Modality           | Mean Threshold | Std. Deviation | Median Threshold |
|--------------------|----------------|----------------|------------------|
| <b>Bending</b>     | 0.85           | 0.02           | 0.86             |
| <b>Compression</b> | 0.33           | 0.01           | 0.33             |

Table 4: Statistical Summary of Failure Threshold Percentages for Bending and Compression Modalities.

### 7.4 Model Assessment

The results of the model assessment, as detailed in Tables 2 and 3, are synthesized and compared against the empirical failure thresholds in Table 5.

| Modality           | Cluster    | Thres. | Med. Earliness (%) |               | $\Delta$ Threshold |         |
|--------------------|------------|--------|--------------------|---------------|--------------------|---------|
|                    |            |        | XGB                | LSTM          | XGB                | LSTM    |
| <b>Bending</b>     | Cluster 0  | 88%    | <b>48.24%</b>      | <b>82.35%</b> | 39.76%             | 5.65%   |
|                    | Cluster 1  | 83%    | <b>72.10%</b>      | <b>51.85%</b> | 10.90%             | 31.15%  |
|                    | Cluster 2  | 84%    | <b>46.17%</b>      | <b>81.97%</b> | 37.83%             | 2.03%   |
|                    | Cluster 3  | 88%    | <b>68.13%</b>      | <b>4.35%</b>  | 19.87%             | 83.65%  |
|                    | No Cluster | 85%    | <b>44.42%</b>      | <b>86.89%</b> | 40.58%             | -1.89%  |
| <b>Compression</b> | Cluster 0  | 33%    | 45.83%             | 93.00%        | -12.83%            | -60.00% |
|                    | Cluster 1  | 34%    | <b>25.27%</b>      | 82.40%        | 8.73%              | -48.40% |
|                    | Cluster 2  | 32%    | 33.16%             | 42.41%        | -1.16%             | -10.41% |
|                    | No Cluster | 33%    | <b>20.30%</b>      | 47.40%        | 12.70%             | -14.40% |

Table 5: Comparison of Median Earliness and Threshold Deviations between XGBoost and LSTM. Bold values indicate successful early predictions (before the threshold).

The introduction of dynamic thresholds significantly reframes the interpretation of the model’s performance. In the **Bending** modality, although the absolute earliness values initially appeared high, the actual failure points occur late in the experimental sequence (approx. 85%). Consequently, achieving a median earliness of roughly one than 50%—as seen with XGBoost—represents a highly positive outcome, providing a substantial predictive lead time. Excluding Cluster 1 and considering the noise in the LSTM results for Cluster 3, XGBoost emerges as the most robust model, consistently delivering accurate predictions well in advance of the structural failure.

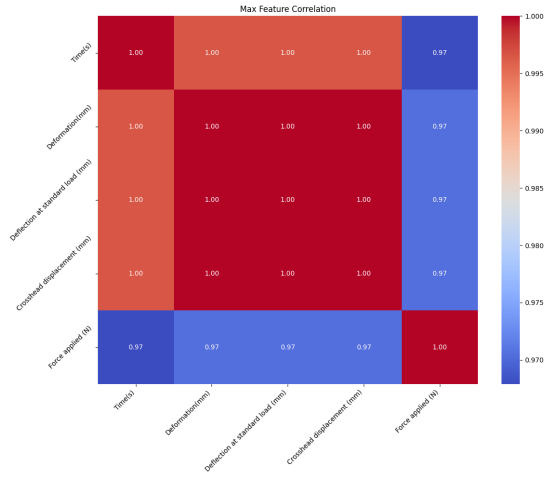
Conversely, the **Compression** results present a more challenging scenario. Since failure occurs much earlier (approx. 33% of the series), effective early prediction is harder to achieve. Only XGBoost in Cluster 1 successfully predicts the failure before the threshold, albeit with a narrower margin (25.27% earliness against a 34% threshold). However, on the complete dataset (No Cluster), XGBoost maintains a strong performance with a positive delta of 12.7%.

In conclusion, the adoption of dynamic thresholds redefines the role of clustering. Clusters have proven essential for analyzing medoid-based thresholds and establishing a reliable mean failure threshold for the entire dataset. This methodology provides a clearer perspective on the results, highlighting superior performance in the Bending tests and confirming the overall higher efficacy of the XGBoost architecture.

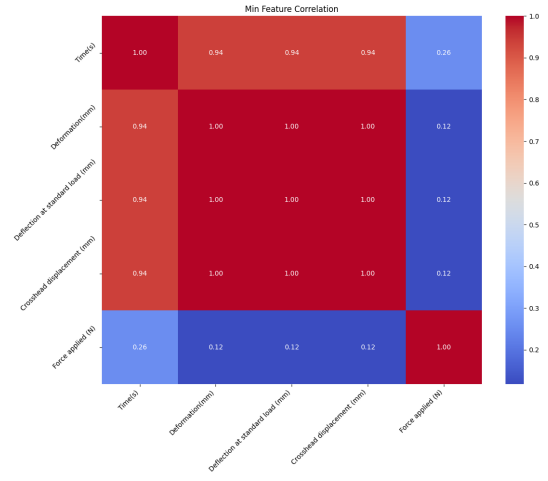
## References

- [1] V. D. Sagias, D. G. Papageorgiou, A. Pappas, and C. I. Stergiou, “Mechanical properties of am polymer specimens under bending stress,” *Applied Mechanics*, vol. 6, no. 2, 2025. [Online]. Available: <https://www.mdpi.com/2673-3161/6/2/25>
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- [3] A. Pellicani, G. Pio, and M. Ceci, “Carrot: Simultaneous prediction of anomalies from groups of correlated cryptocurrency trends,” *Expert Systems with Applications*, vol. 260, p. 125457, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417424023248>

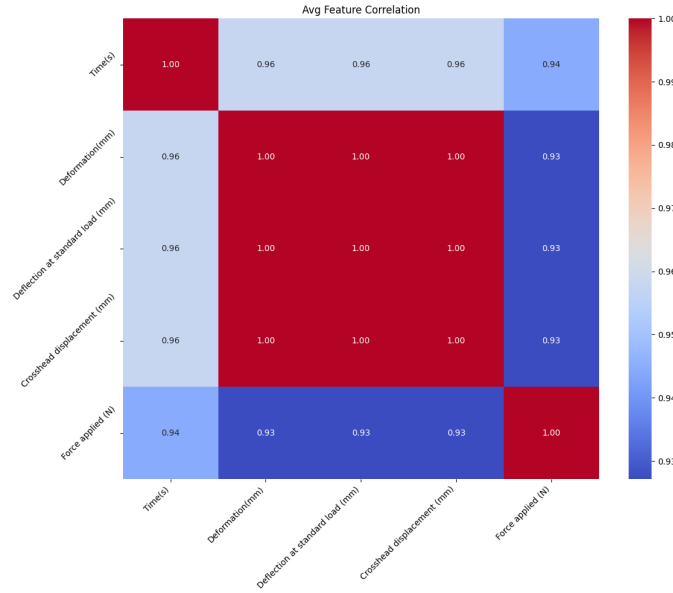
# A Experimental Data visualization



(a) Maximum Feature correlation

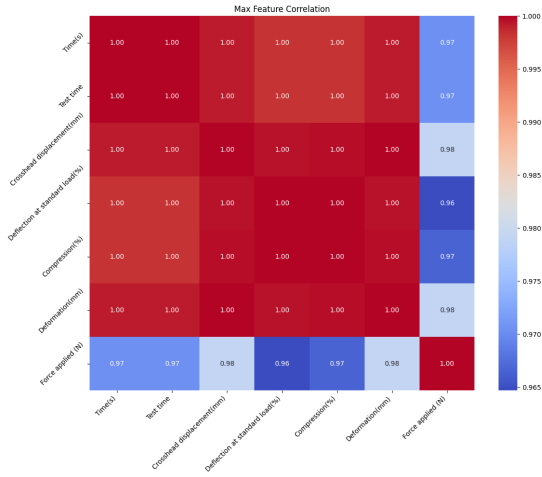


(b) Minimum Feature correlation

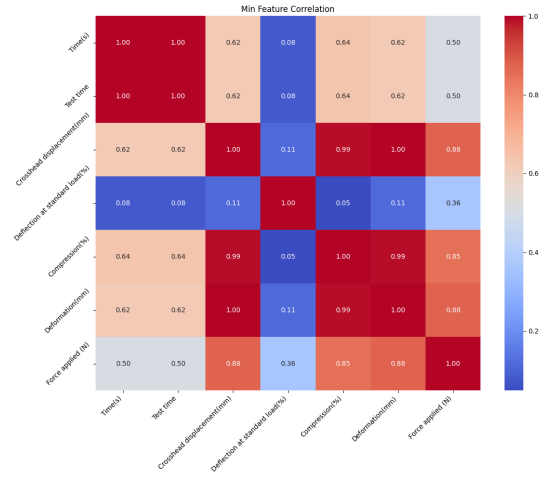


(c) Average feature correlation.

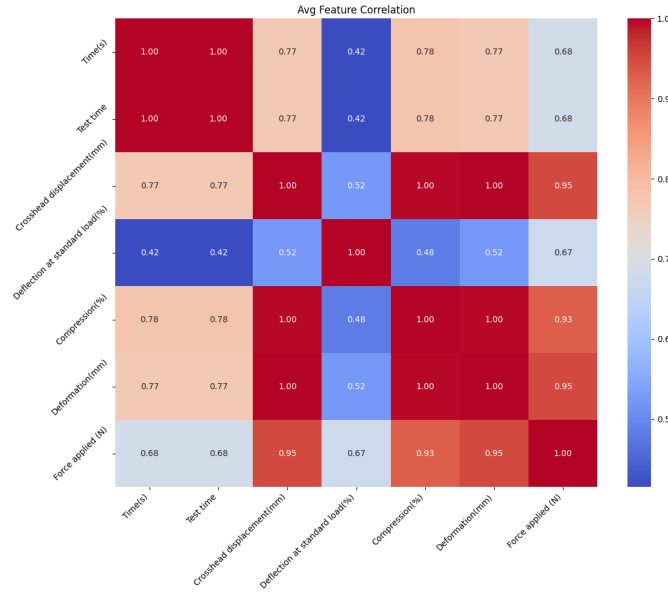
Figure 1: Bending input feature correlation



(a) Maximum Feature correlation

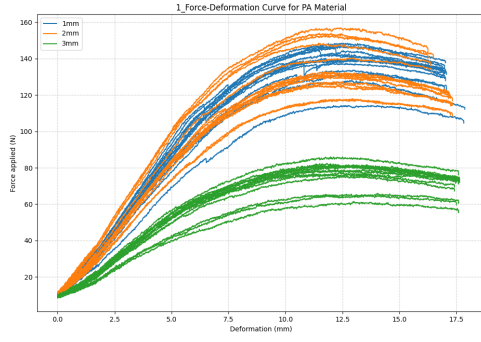


(b) Minimum Feature correlation

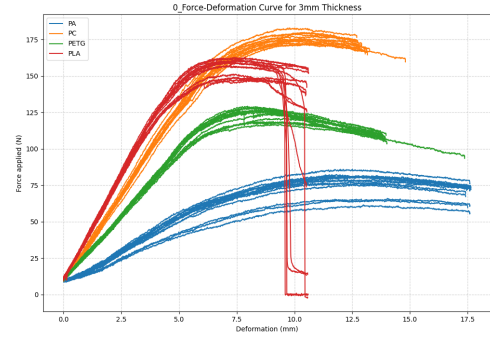


(c) Average feature correlation.

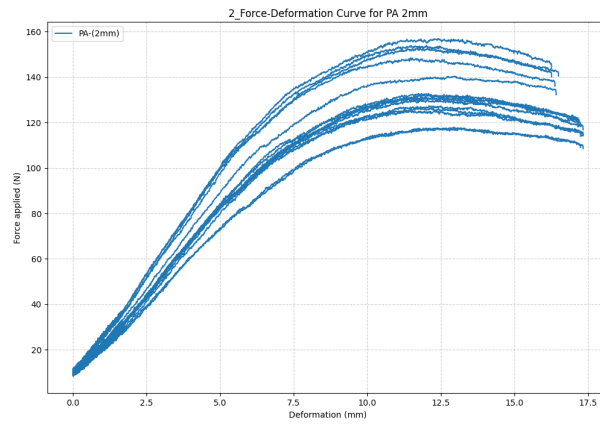
Figure 2: Bending input feature correlation



(a) Stress-strain trajectories grouped by material type.

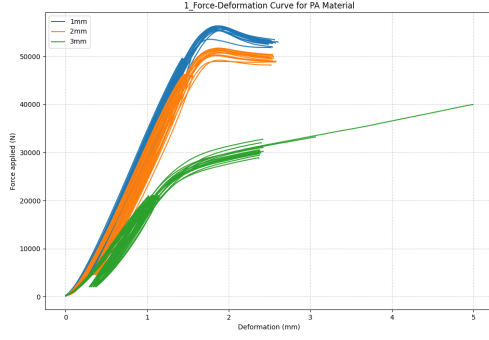


(b) Stress-strain trajectories grouped by material type.

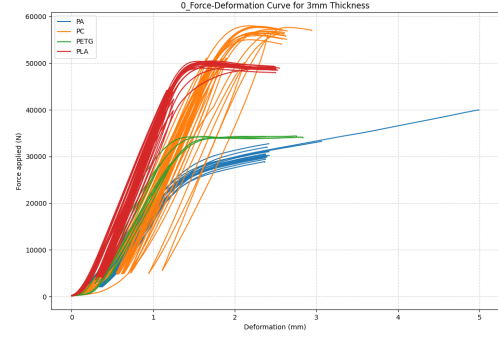


(c) Stress-strain trajectories grouped by material type and thickness.

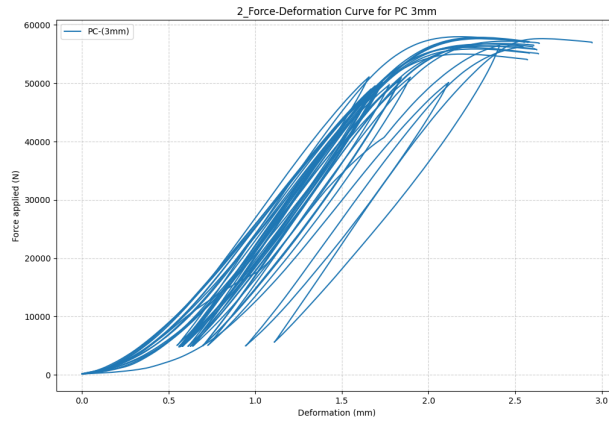
Figure 3: Stress-strain curves Bending



(a) Stress-strain trajectories grouped by material type.



(b) Stress-strain trajectories grouped by material type.



(c) Stress-strain trajectories grouped by material type and thickness.

Figure 4: Stress-strain curves Compression

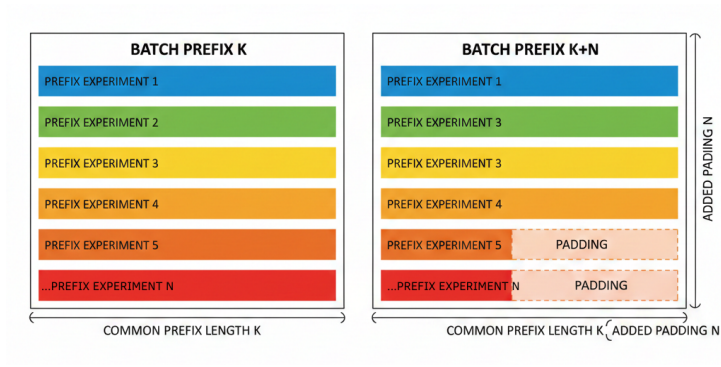
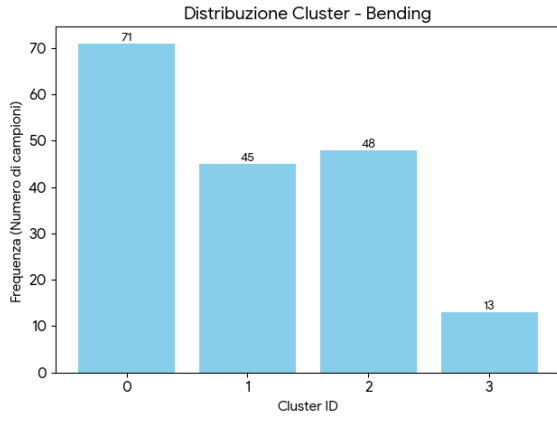
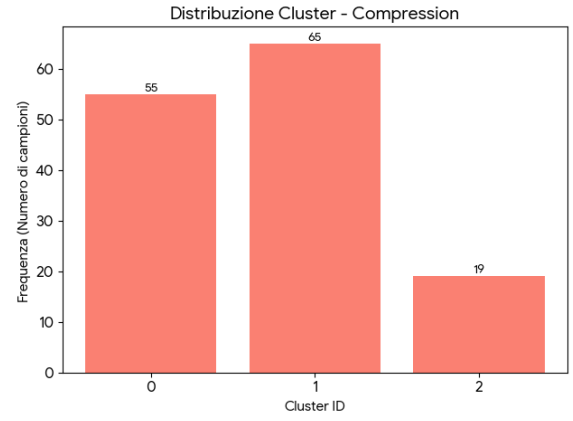


Figure 5: Schematic representation of sequential data batching.

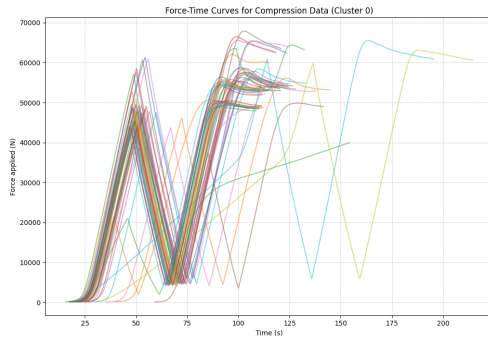


(a) Cluster distribution Bending.

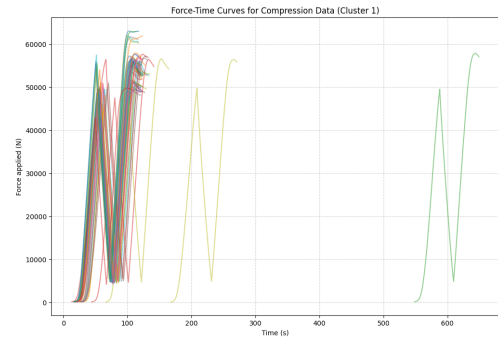


(b) Cluster distribution Compression

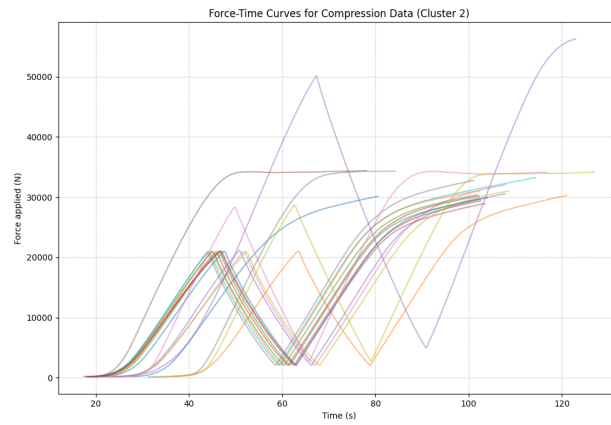
Figure 6: Cluster distribution of both groups



(a) Force Curve on cluster 0

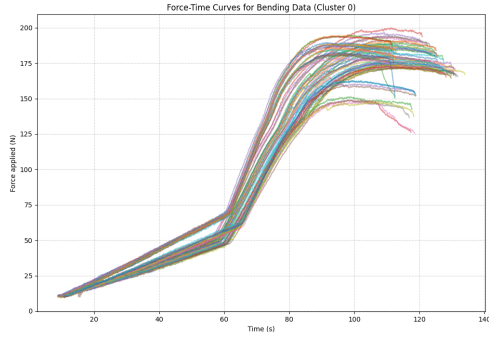


(b) Force Curve on cluster 1

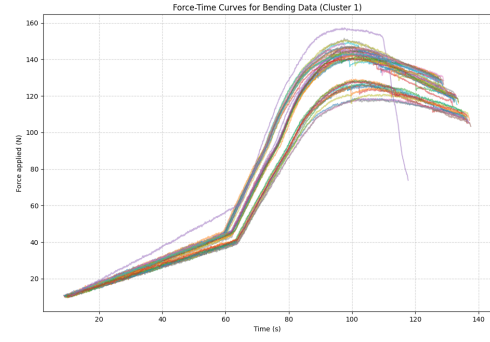


(c) Force Curve on cluster 2

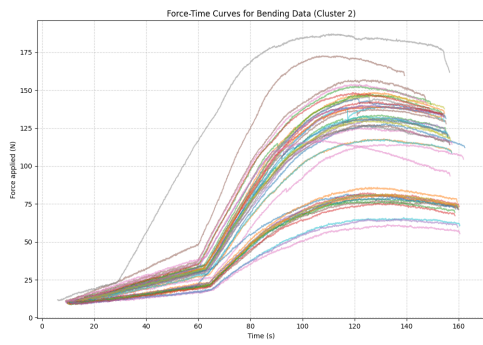
Figure 7: Force Curves for compression



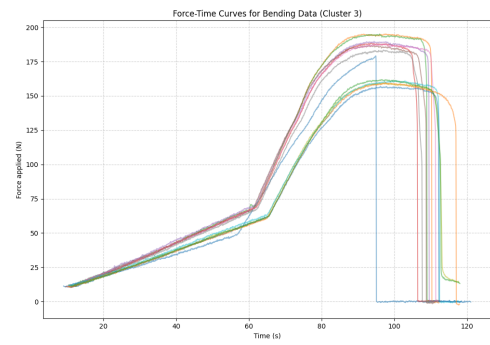
(a) Force Curve on cluster 0



(b) Force Curve on cluster 1

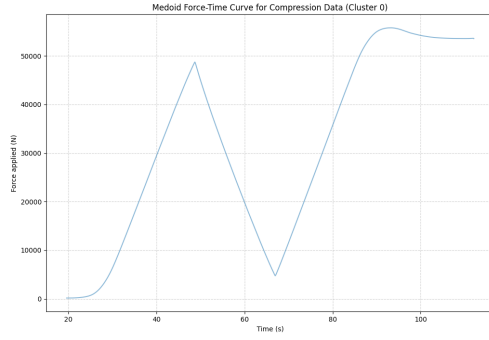


(c) Force Curve on cluster 2

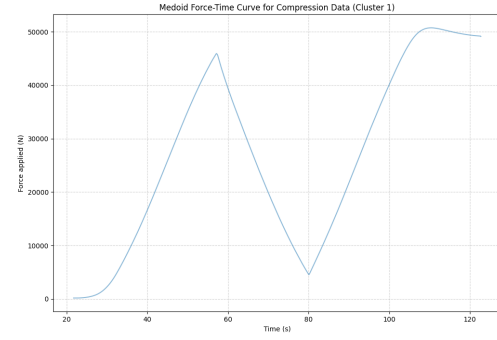


(d) Force Curve on cluster 3

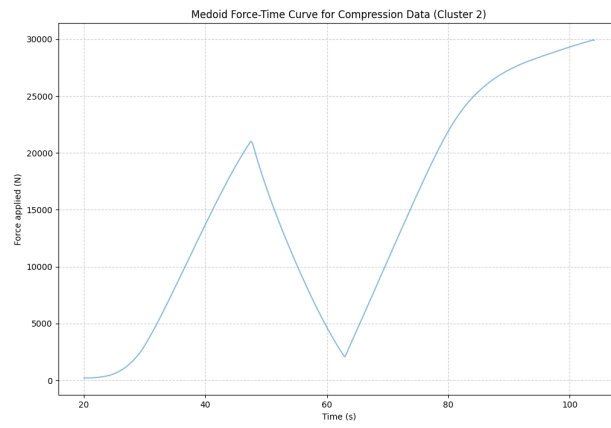
Figure 8: Force Curves for Bending



(a) Force Curve on the medoid of Cluster 0

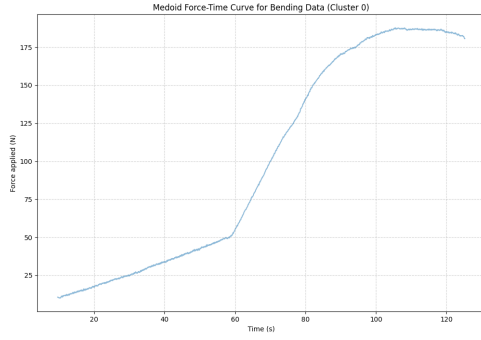


(b) Force Curve on the medoid of cluster 1

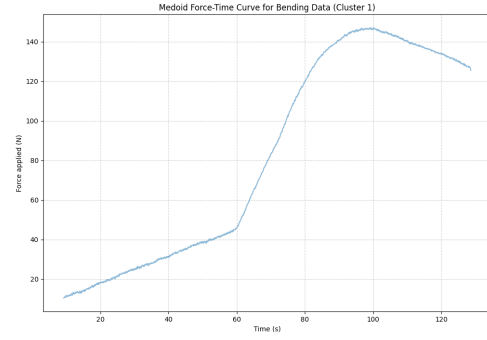


(c) Force Curve on the medoid of cluster 2

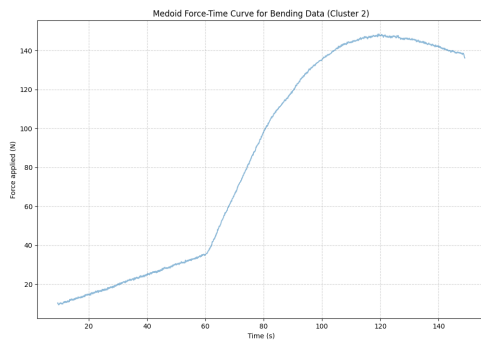
Figure 9: Medoids Force Curves for compression



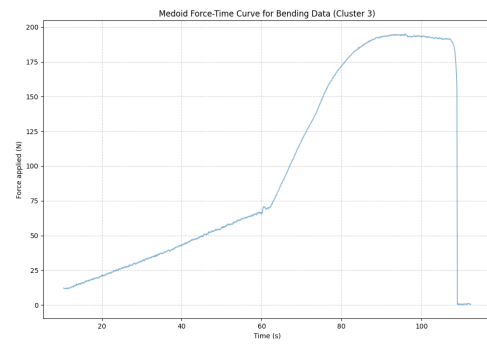
(a) Force Curve on the medoid of cluster 0



(b) Force Curve on the medoid of cluster 1



(c) Force Curve on the medoid of cluster 2



(d) Force Curve on the medoid of cluster 3

Figure 10: Medoid Force Curves for Bending